

Spotlight Reflex

Minimal-Shot Autonomous Driving via Geometric World-State Reasoning

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Agenda

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Motivation & Claim

Who I Am & What I Built

Alba Maria Tellez Fernandez · Lead AI,
robotics logistics startup

Question behind this project:

- Can a compact geometric representation replace learned scene priors?
- Does *“avoid occupied space, preserve route”* generalise to geometry never seen in training?

Two independent tracks — both from scratch:

- **Spotlight Reflex** — closed-loop simulation policy
- **WOD-E2E harness** — trajectory selector on 479 real Waymo frames

Main Claim

A driving policy that generalises to long-tail scenarios by reasoning from **occupancy, clearance, and route geometry** — not by memorising scenario labels or imitating average trajectories.

This is NOT

- A production AV stack
- A strict zero-shot WOD-E2E claim
- A trained camera-perception system

The Problem — and the Failure

Imitation learning minimises:

$$\min_f \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f(x), y)]$$

Works on the common driving distribution $\mathcal{D}$.

Fails when $P_{\text{test}} \neq P_{\text{train}}$ — rare geometry, no fallback reasoning.

WOD-E2E targets **11 scenario types**

occurring $<0.03\%$ of driving time: wrong-way actors, animals, debris, construction, cut-ins, cyclists.

Baseline: trajectory extrapolation only, no world-state reasoning. Drives directly into the actor.



Wrong-way actor, night. Baseline has no awareness the path is occupied.

Policy Architecture

Core Idea: Reason from Geometry, Not Memory

*"A trajectory that avoids occupied space and respects route geometry is correct **regardless of what the obstacle is called.**"*

Formal invariance: let T be any relabelling of object identities, scenario names, or cluster tags that leaves occupancy G and route R unchanged.

$$M(s) = M(T(s)) \quad \forall s, \forall T$$

"sheep" $\rightarrow$ "debris" changes nothing.

Regression-tested: a test suite relabels all objects in every scene and asserts maneuver, score, and reason are identical. Passes.



*Same scenario, Spotlight Reflex: selects **evasive_right**, clears actor, returns to lane. Geometry resolved the decision.*

Six World-State Scalars

All signals computed from raw obstacle positions and route geometry. **Nothing is learned; nothing is scene-specific.**

Signal	Definition	Range
obstacle_pressure	$p = \text{clip}\left(\frac{10-d_{\min}}{10}, 0, 1\right)$	$[0, 1]$
route_blockage	fraction of forward corridor occupied	$[0, 1]$
corridor_blocked	hard blockage within stopping distance	bool
left_clearance	free lateral space to the left (m)	≥ 0
right_clearance	free lateral space to the right (m)	≥ 0
preferred_escape_side	$\arg \max(\text{left}, \text{right})$	L / R

$d_{\min}$ = distance to nearest obstacle. Pressure = 1.0 at contact, 0 beyond 10 m.

Nine Maneuver Candidates

Each candidate: **20-point, 5 s trajectory at 4 Hz** via Hermite smoothstep. **Evasive maneuvers** apply the lateral offset early (first third) for fast separation.

Maneuver	Intent	Speed scale	Lateral offset
stop	Emergency stop	0.00×	0.0 m
crawl	Very slow progress	0.15×	0.0 m
maintain	Hold current speed	1.00×	0.0 m
slow_yield	Decelerate & yield	0.50×	0.0 m
nudge_left	Minor left shift	0.80×	+0.9 m
nudge_right	Minor right shift	0.80×	−0.9 m
evasive_left	Strong left evasion	0.70×	+2.2 m
evasive_right	Strong right evasion	0.70×	−2.2 m
lane_recover	Return to centre	0.90×	0.0 m

Trust-Region Scoring & Policy Architecture

Speed-scaled tolerance:

$$s(v) = \text{clip}\left(\frac{v-1.4}{9.6}, 0, 1\right) \times 0.5 + 0.5$$

Horizon	Lateral	Longitudinal
3 s	$1.0 \times s(v) \text{ m}$	$4.0 \times s(v) \text{ m}$
5 s	$1.8 \times s(v) \text{ m}$	$7.2 \times s(v) \text{ m}$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s} + 0.5 \cdot \max_r \text{score}_{5s}$$

Scores ≤ 50 = unsafe floor.

How candidates are selected:

1. Scene $\rightarrow$ compute 6 world-state scalars
2. Reference rule engine maps scalars to expected maneuvers + scores
3. Each of the 9 candidates is scored against references at 3 s and 5 s
4. Highest combined score wins

Every step emits a full `SpotlightSelection` log: maneuver, 3 s/5 s scores, winning reference, decision reason, all scalar values. In `AlpaSim`: serialised as `reasoning_text` JSON.

Policy Architecture — Decision Flow



6 Scalars: pressure, blockage, corridor blocked, left/right clearance, preferred escape side — all from raw geometry.

Ref Rules: world-state thresholds → expected maneuver + score (80–94 scale; ≤ 50 = unsafe).

Scorer: 3 s + 5 s horizons, speed-scaled tolerance. Full log emitted every step.

Simulation Results

The Simulator & Compositional OOD

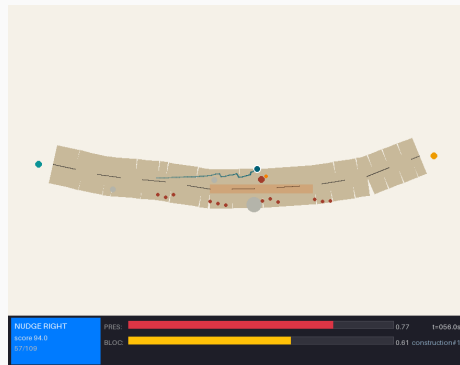
Built from scratch — no Waymo or AlpaSim dependency.

- Closed-loop 2D engine: ego + actors step simultaneously
- All 11 WOD-E2E scenario clusters implemented
- 8 actor behaviour models (linear, cut-in, swerve, erratic, wrong-way. . .)
- COMPASS composite benchmark (threshold 7.0/10)

Compositional OOD:

$$P(\text{topology, hazard}) = P(\text{topology}) \times P(\text{hazard})$$

8 topologies $\times$ 11 hazards $\times$ 4 weather modes,
sampled independently — a construction hazard can
appear in a roundabout in fog.



Construction zone — 5.2m corridor, cone field, lane closure. Policy: `nudge_right`.

Simulation Results

Primary benchmark — 350 rollouts, seeds 1–10, deterministic:

Suite	Runs	Coll.	Pass
WOD-style (11 clusters)	110	0	100%
Compositional OOD	60	0	100%
Adversarial (2–3 hazards)	60	0	100%
Hidden holdout	60	0	100%
Gauntlet (4h, 3.5 m)	60	0	60%
Total	350	0	93.1%

COMPASS 9.137/10 · CI [0.0, 0.0053]

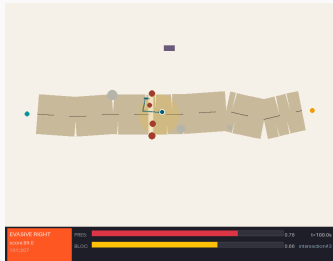
Gauntlet — 420 matched scenarios:

Policy	Pass	Collision
Baseline	2.1%	20.5%
Spotlight Reflex	57.6%	7.9%

27× more passes · 2.6× fewer collisions.

60% gauntlet = failure boundary (4 hazards in 3.5 m).

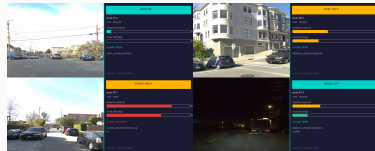
Scenarios & AlpaSim Transfer



Intersection stress — two crossing actors, 207 steps. Zero collisions.



*FOD — debris, primary lane blocked. *nudge_right*.*



AlpaSim — real Waymo camera + adapter output.

4-channel adapter:

route, brightness, ego dynamics, hazards.

Zero AlpaSim imports in the policy.

30 scenes:

0.0 collision at fault

0.42 m dist. to GT traj.

0.37 any-collision (others)

WOD-E2E Benchmark

WOD-E2E: Benchmark & Selector Pipeline

Waymo Open Dataset End-to-End:

- 479 validation frames from real Waymo recordings
- Human raters: pairwise preferences between candidate trajectories
- **RFS** = how often the selected trajectory was preferred. Higher = better.
- **5-fold segment-grouped CV** — frames from the same driving segment stay together. Fold std ≈ 0.19 , CI $\approx \pm 0.17$.

Two backends — not interchangeable

Local: 7.131 · Official Waymo: 7.022. 2-fold $\neq$
5-fold — not comparable.

Candidate sources:

- **Kinematic:** const-vel, const-accel, heading change, stop
- **Ridge learned:** 31 trajectory features; target = gain vs baseline
- **Temporal:** ego-history trends; wins on 49% of frames

Selection:

HGB gate ranker (~ 137 features, 5-fold) picks best candidate.

Direct policy overrides on $\sim 4\%$ of frames when confidence high.

Frames $\rightarrow \{\text{Kinematic} \text{ — Ridge — Temporal}\} \rightarrow \text{HGB Gate} \rightarrow \text{Direct Policy} \rightarrow \text{Output}$

Visual Embeddings & WOD-E2E Results

Do image embeddings help the ranker?

Cosmos (NVIDIA): video autoencoder — 64-dim continuous latents.

InternVLA (Shanghai AI Lab): ViT+LLM — 128-dim ViT output. Both failed under a linear (ridge) head. Non-linear model required.

Config	Head	RFS
No embedding	Ridge	7.803
Cosmos 64d	Ridge	≈7.803
InternVLA 128d	Ridge	≈7.803
RFF gate	non-lin.	7.834
Cosmos 64d	GPU MLP	7.845

Embeddings carry signal — only a non-linear model extracts it.

Full results — 5-fold segment-grouped CV:

Selector	Folds	RFS
Official Waymo baseline	—	7.022
Local CV baseline	—	7.131
Gate only	5	7.803
RFF direct policy	5	7.834
GPU MLP + Cosmos 64d	5	7.845
HGB Optuna peak	2 only	7.880
Oracle (perfect selector)	5	9.264

Champion: h=64 MLP, lr=0.003866, 4.2% of frames, precision **0.60**

Oracle gap: 1.419 RFS — right candidate exists, selector can't identify it without visual scene information.

Honest Failures & Next Steps

What didn't work:

Failure	Root cause
No camera perception	Sees only ego velocity, speed, accel, route intent. Cannot see debris or signals.
Embeddings under linear head	Cosmos/InternVLA as ridge features — no gain. Signal requires a non-linear model.
HGB Optuna 2-fold	7.880 under 2-fold; failed 5-fold. Optuna exploited fold structure.
Turn calibration	GO_RIGHT regret 2.063. Only 29 training frames for that intent.

Next steps:

1. **Task-aligned encoder** (highest impact)
Encoder fine-tuned on preference triplets (frame, win, loss). Expected: 0.5–1.5 RFS. Closes most of the 1.419 oracle gap.
2. **Turn calibration**
Stratified reweighting by intent class.
GO_LEFT: 1.753→1.614 already. GO_RIGHT (2.063) next.
3. **Test-set submission**
Pipeline validated; 1,505 frames listed.
Missing: test TFRecords (Waymo sign-in gated).
4. **Physical deployment**
Obstacle sensors + route command →
Spotlight Reflex today. AlpaSim adapter is the real-sensor bridge template.

Summary

Core claim	Geometric reasoning from 6 scalars generalises without training data for the scenario
Invariance	$M(s) = M(T(s))$ — regression-tested: relabelling all objects $\rightarrow$ identical decisions
Simulator	Built from scratch · 11 WOD clusters · compositional OOD · COM-PASS 9.137/10
Gauntlet	57.6% pass · 7.9% collision vs baseline 2.1% / 20.5% · 27 $\times$ improvement
WOD-E2E champion	7.845 RFS (5-fold, GPU MLP + Cosmos 64d) · +0.714 vs local baseline
Oracle gap	1.419 RFS — candidate pool is good; discriminator is the bottleneck
AlpaSim	Same policy, sensor-realistic · 0.0 collision at fault · 0.42 m to GT
Next step	Task-aligned encoder fine-tuned on preference triplets

Generalises from geometry, not memorised trajectories. The 1.419 RFS oracle gap is recoverable with a preference-aligned visual encoder.

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`docs/simulation.md · docs/wod-e2e-system-walkthrough.md`

Appendix: Reference Rule Engine

Maps world-state threshold conditions to pseudo-rater references (name + score).

World condition	References (score)
Low pressure (< 0.25), low uncertainty (< 0.45), open corridor (> 0.35)	maintain (92), lane centre (88)
Moderate pressure (> 0.20)	slow_yield (86), best-side nudge (94), evasive (89)
High uncertainty (> 0.55)	crawl (90), stop (82)
Lane offset > 0.3 m and speed > 0.5 m/s	lane_recover (93)
Default fallback	maintain (80), slow_yield (76)

Scores 80–94 on a scale where ≤ 50 = unsafe rating.

Appendix: Calibration Bias Audit

Regret = oracle RFS – selected RFS. Where does the error live?

Slice	n	Selected	Oracle	Regret
GO_STRAIGHT	427	7.959	9.336	1.377
GO_LEFT	23	7.107	8.721	1.614
GO_RIGHT	29	6.574	8.638	2.063
speed:fast	44	7.957	9.436	1.479
speed:slow	133	7.564	9.196	1.632
speed:urban	136	8.132	9.416	1.285

GO_RIGHT ($n=29$) and GO_LEFT ($n=23$) are rare — ranker is calibrated to GO_STRAIGHT (89% of frames). GO_LEFT improved 1.753→1.614 with targeted changes. **Fix:** stratified loss reweighting by intent class.

Appendix: RFF Derivation & Optuna TPE

Random Fourier Features — approximate Gaussian kernel without $N \times N$ matrix:

$$\mathbf{W} \sim \mathcal{N}(0, \sigma^{-2}I), \quad D=512, \quad \sigma=7.858$$

$$\varphi(\mathbf{x}) = \sqrt{\frac{2}{D}} \cos(\mathbf{W}^\top \tilde{\mathbf{x}} + \mathbf{b})$$

$$\mathbb{E}[\varphi^\top \varphi'] = \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2\sigma^2}\right)$$

Only 512-dim ridge weights are learned. RFS: 7.834, precision 0.41.

Optuna TPE — Bayesian optimisation:

$$\ell(x) = P(x \mid \text{top-}p)$$

$$g(x) = P(x \mid \text{bottom})$$

$$x_{\text{next}} = \arg \max_x \frac{\ell(x)}{g(x)}$$

GPU MLP search (20 trials):

- Best: trial_0004 — $h=64$, $\text{lr}=0.003866 \rightarrow$ **7.845**
- Mean 20 trials: 7.769
- 1/20 beat the RFF baseline (7.834)
- With 479 frames + 5 folds, Optuna can exploit fold patterns